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Inorganica Chimica Acta 296 (1999) 1–11

**Inorganica
Chimica Acta**

Review

Copper-mediated 1,6-, 1,8-, 1,10- and 1,12-addition and 1,5-substitution reactions in organic synthesis

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Received 15 June 1999; accepted 3 September 1999

Abstract

Of all the organometallic reagents currently used to form carbon–carbon bonds, organocopper compounds rank amongst the most important. Classical applications of these reagents involve carbocupration, 1,4-Michael addition and 1,3-(S_N2′)-substitution reactions. Recently, the scope of organocopper chemistry has been broadened significantly by the development of 1,6-, 1,8-, 1,10-, and 1,12-addition reactions, as well as of 1,5-substitution reactions of multiply unsaturated substrates which often contain at least one triple bond. These new transformations not only proceed with high regio- and stereoselectivity and can therefore be utilized in target-oriented synthesis, e.g. of functionalized allenes, but they also provide new information about the mechanism of copper-mediated carbon–carbon bond forming reactions. In this account some of the more recent important developments in this area are discussed. © 1999 Elsevier Science S.A. All rights reserved.

Keywords: Organocopper reagents; Michael additions; Substitutions; Allenes; Regioselectivity; Stereoselectivity

1. Introduction

Although the element copper has been known since ancient times and has been used for a long time in established reactions, such as the Sandmeyer reaction, the Ullmann coupling and coupling reactions of alkynes, the deliberate utilization of organocopper reagents in organic synthesis started only about 40 years ago. Based on the seminal work of Gilman who carried out the first investigations of organocopper compounds RCu [1] and lithium diorganocuprates R₂CuLi [2], it was quickly established by House et al. and Corey et al. that the latter are the reagents of choice for carbon–carbon bond formation since they are highly reactive towards a wide variety of electrophiles (even today, cuprates of the type R₂CuLi are referred to as Gilman reagents in honor of his pioneering work). Thus, organocuprates undergo substitution reactions on many

saturated (haloalkanes, acid chlorides, oxiranes) and unsaturated electrophiles (allylic and propargylic derivatives), as well as 1,4-additions to α,β -unsaturated carbonyl compounds and carbocuprations of nonactivated alkynes [3].

Nowadays, *reactivity* is only one of the necessary qualities of a successful class of reagents, the others being *efficiency* and *selectivity*. The major advantage of organocopper compounds is that each of these features can be controlled by ‘tuning’ of the reagent, making them highly useful for the total synthesis of complex target molecules like natural products, pharmaceuticals and chiral auxiliaries [3]. For example, many different combinations of copper(I) salts and organometallic compounds can be employed for the preparation of organocopper reagents with different composition and reactivity; particularly commonly used are CuI, CuBr·Me₂S and CuCN as copper source and organolithium, Grignard and organozinc reagents as transmetalating agents (the latter also allows for the introduction of functional groups into cuprates [4]). Diethyl ether, THF, and dimethyl sulfide are suitable

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solvents for the preparation and reactions of organocopper reagents RCu and cuprates R_2CuLi . Since these solvents also serve as ligands [5], a change of the solvent strongly affects the reactivity [3]. By variation of the stoichiometry of copper salt and organometallic reagent, different cuprates can be obtained; for example, the reaction of 1 equiv. of CuCN with 1 equiv. of RLi leads to the so-called lower-order cyanocuprates RCu(CN)Li wherein the cyanide is bound to the copper center [6,7] (see [7a–c] for solid-state structures of cyanocuprates); after addition of a second equiv. of RLi , cyano-Gilman reagents $\text{R}_2\text{CuLi} \cdot \text{LiCN}$ are formed which do not contain a copper-bound cyanide ion in their thermodynamically most stable form [7]. The former are an example of mixed cuprates of the type $\text{R}_t\text{R}_r\text{CuLi}$, in which R_t is an easily transferred ligand and R_r is a tightly held, ‘residual’, nontransferable ‘dummy ligand’. These reagents are particularly useful with ‘valuable’ substituents R_t since generally only one of the two R groups in a cuprate of the stoichiometry R_2CuLi can be introduced into a substrate. Besides cyanide, commonly used dummy ligands are cheap and easily accessible alkynyl, thienyl, alkoxy, thiolato, and amido groups [3].

Recent developments in organocopper chemistry focus on the discovery of new catalytic and/or enantioselective versions of copper-mediated transformations. Although copper-catalyzed 1,4-addition reactions of Grignard reagents to enones were reported by Kharasch and Tawney already in 1941 [8], the search for new copper-catalyzed reactions has continued ever since, because copper is a problematic element in environmental terms [9,10]. A particularly promising approach makes use of the conversion of terminal alkynes into alkenylzirconocenes by hydrozirconation with the Schwartz reagent $\text{Cp}_2\text{Zr(H)Cl}$; these can then be used in copper-catalyzed 1,4-additions to enones [10]. Enantioselective 1,4-additions have been carried out with limited success by using chirally modified mixed cuprates $\text{RCu(L}^*\text{)Li}$ wherein the chiral nontransferable ligand L^* (usually an alcoholate or amide) controls the stereochemical course of the transfer of the group R to the substrate [11]. The well known dynamic behavior of organocopper species in solution has presented difficulties inasmuch as the only very modest enantioselectivities often observed with chiral cuprates $\text{RCu(L}^*\text{)Li}$ may well be due to equilibria with the achiral, but more reactive homocuprates R_2CuLi which form racemic 1,4-adducts. Recently, these problems have been overcome by the discovery of a new class of chiral ligands wherein binaphthol and a chiral amine are bridged via a phosphorus center, allowing highly enantioselective copper-catalyzed 1,4-addition reactions of organozinc compounds to selected enones [12].

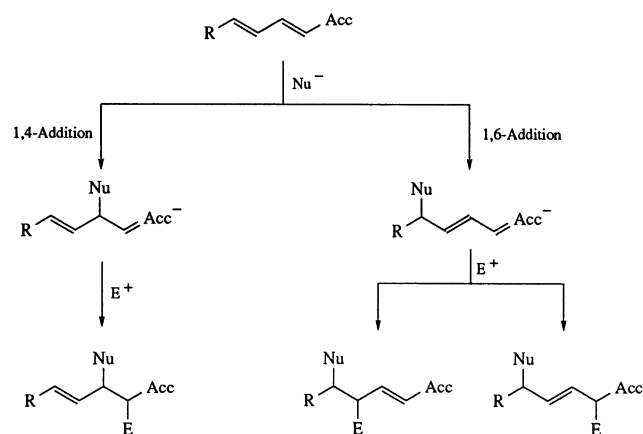
All recent investigations described above have led to the development of highly sophisticated organocopper

reagents and new protocols for conducting their reactions. In terms of the substrates, however, little has changed until recently, i.e. the usual 1,4-Michael addition and 1,3-($\text{S}_{\text{N}}2'$)-substitution reactions of simple unsaturated substrates were carried out. Only in the last 10 years or so, reactions involving ambident substrates with extended multiple bond systems (in other words, with two or more reactive positions) were examined and found to take place with high regio- and stereoselectivities. Indeed, this is true not only for substitutions (1,5-substitutions), but also for additions (1,6- and 1,8-additions, etc.), in particular when the substrate contains at least one triple bond besides one or more conjugated double bonds. These unusual reaction types not only open up novel entries to interesting target molecules but also provide deeper insight into the mechanisms of copper-mediated bond formation [30,13].

2. Copper-mediated addition reactions to extended Michael acceptors

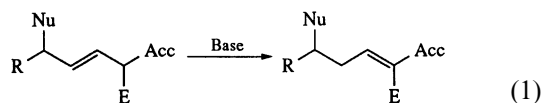
2.1. Acceptor-substituted dienes

Due to their ambident character, acceptor-substituted dienes can provide several isomeric products in copper-mediated Michael additions, making it particularly important to control the regio- and stereoselectivity of these transformations (Scheme 1). Besides direct nucleophilic attack at the acceptor group, an activated diene may undergo a 1,4- or 1,6-addition; in the latter case, capture of the ambident enolate with a soft electrophile can also take place at two different positions. Thus, the nucleophilic addition can produce three regioisomeric alkenes which may be formed as *E/Z* isomers. Depending on the nature of nucleophile and electrophile, the adducts can also contain one or two centers of chirality. An additional complication arises from the fact that

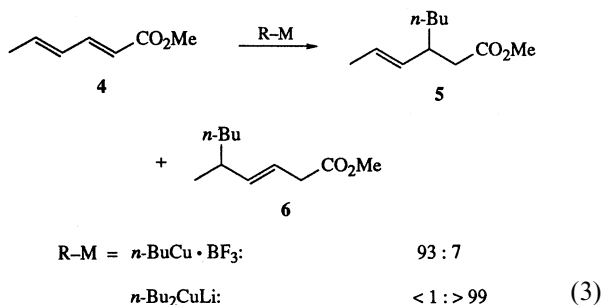
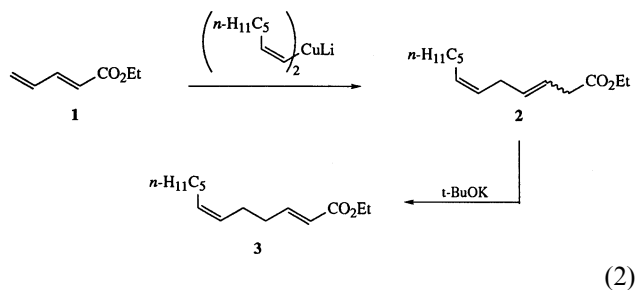


Scheme 1.

β,γ -unsaturated carbonyl compounds (and other acceptor-substituted alkenes of this type) are readily isomerized to the thermodynamically more stable conjugated isomers under basic conditions (Eq. (1)).

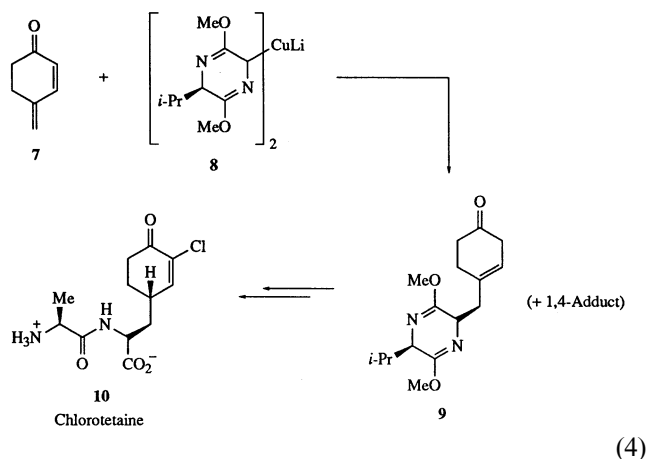


The first example of a cuprate addition to an acceptor-substituted diene was reported by Näf et al. [14] who used lithium di-(*Z*)-1-heptenylcuprate in a Michael addition to ethyl-2,4-pentadienoate (**1**). The reaction proceeded highly regioselectively to furnish a 1:1 mixture of isomeric 1,6-adducts **2** which was converted into the Bartlett pear constituent ethyl (2*E*,6*Z*)-2,6-dodecadienoate (**3**) by isomerization with base (Eq. (2)). Similar reactions described by many other groups [15] always provided 1,6-addition products exclusively, i.e. there was no indication of the possibility of 1,4-cuprate additions to activated dienes. However, Yamamoto et al. [3f,16] showed in their seminal contributions that this is indeed feasible: while the reaction of methyl sorbate (**4**) with the reagent formed from butylcopper and the Lewis acid boron trifluoride led to the 1,4-adduct **5** as the major product, the Gilman cuprate *n*-Bu₂CuLi provided exclusively the 1,6-addition product **6** (Eq. (3)). The synthetically very useful BF₃-activated organocopper compounds (oftentimes written as 'RCu·BF₃') have been named Yamamoto reagents [3f].



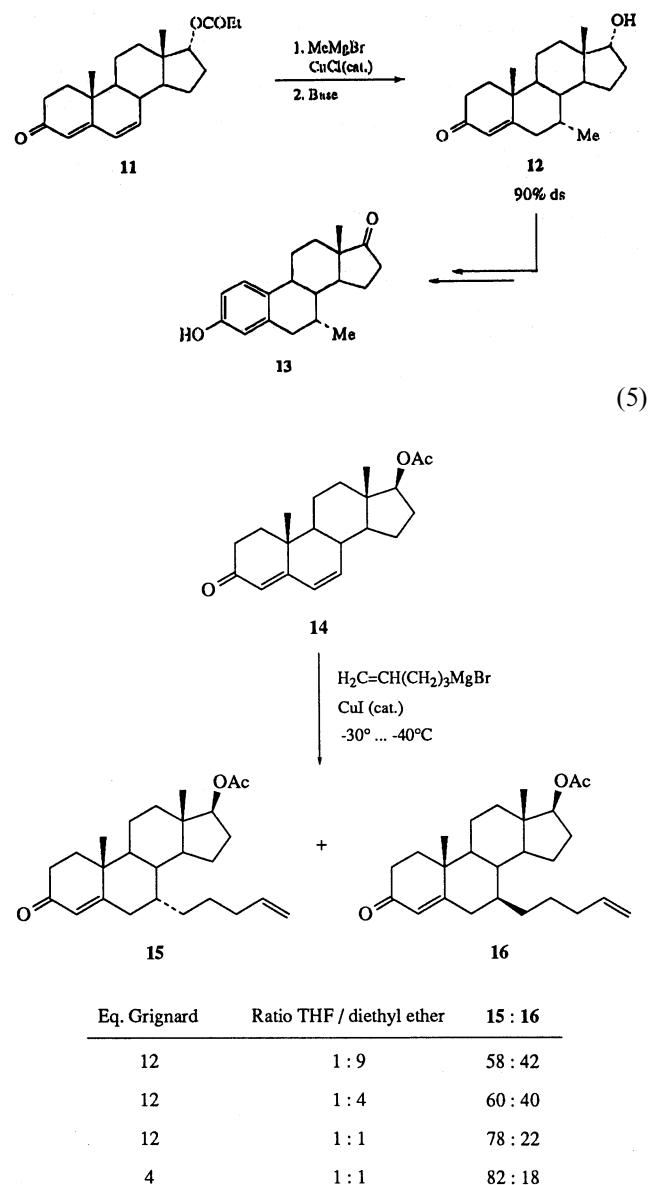
Michael additions of organocopper reagents to acceptor-substituted dienes have been utilized frequently in target-oriented stereoselective synthesis [17]. For example, the cuprate **8** containing a chiral Schöllkopf bislactim ether moiety was used in the first total synthe-

sis of the antimycotic dipeptide chlorotetaine (**10**; Eq. (4)) [17d]. In this case; however, the nucleophilic addition to dienone **7** did not proceed with high regioselectivity since a 63:37 mixture of the 1,6- and 1,4-adduct was obtained. The 1,6-addition product **9** was converted over several steps into diastereo- and enantiomerically pure chlorotetaine (**10**).



Copper-catalyzed Michael additions of Grignard reagents to activated dienes were reported even earlier than those of (stoichiometric) organocuprates. The favorite substrates were steroids with a dienone structure [18], as well as tetrahydro-3H-naphthalen-2-ones, which can be considered as model substrates for doubly unsaturated steroids [19]. The interest in these transformations was prompted by the desire to prepare new, unnatural corticosteroids with possible interesting pharmacological activities. Particularly intriguing are estradiol derivatives bearing an alkyl chain in the 7 α -position since these steroids were found to bind with high affinity and specificity to estrogen receptors; i.e. they are effective anti-estrogenic agents [20] and may therefore be useful for the treatment of mammary tumors (breast cancer) [21]. The obvious way to introduce a carbon group in the 7-position of a steroid backbone is a copper-mediated 1,6-addition to an activated doubly unsaturated $\Delta^{4,6}$ -derivative; indeed, the desired regioselectivity is observed in many cases [18,19,21,22]. Here, the major challenge is control of the diastereoselectivity of the Michael addition since the 7 β -isomers are less effective enzyme inhibitors [21]. In general, a high preference for the addition of methyl nucleophiles from the α -side was observed [18]; for example, Wieland and Auner [18e] reported an α -selectivity of 90% in the copper-catalyzed 1,6-addition of MeMgBr to dienone **11** (Eq. (5)). The product **12** was converted over several steps into 7 α -methylestrone (**13**). By contrast, introduction of longer alkyl chains via copper-promoted 1,6-addition reactions to $\Delta^{4,6}$ -steroids normally gives unsatisfactory $\alpha:\beta$ ratios [21,22]. In some cases, however, the diastereoselectivity could be improved by 'fine-tuning' of the reaction conditions; for

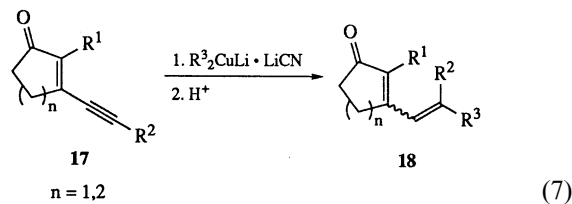
example, the ratio of the epimeric products **15** and **16** in the copper-catalyzed 1,6-addition of 4-pentenylmagnesium bromide to dienone **14** rose from 58:42 to 82:18 upon variation of the amount of nucleophile and solvent composition (Eq. (6)) [22f].



(5)

from the preparative and mechanistic point of view [30,24]. Again, several regioisomeric products can be formed by cuprate addition to these ambident substrates; thus, the addition may take place in a 1,4- or 1,6-fashion, and the ambident allenyl enolate formed in the latter case can trap a soft electrophile to furnish either an allene or a conjugated diene (Scheme 2). Several stereoisomeric products can be obtained in each case, and the possibility of the formation of axial chirality is particularly intriguing. Recent investigations have demonstrated that the regio- and stereoselectivity of both steps can be controlled by the choice of the reactants, in particular by 'fine-tuning' of the organo-copper reagent and the electrophile.

The first copper-mediated additions to enynes with an acceptor group at the triple bond were reported by Hulse [25] who found that 3-alkynyl-2-cycloalkenones **17** react with cuprates at the triple bond in a 1,6-addition and the allenyl enolates are protonated at C-4 with the formation of conjugated dienones **18** as mixtures of *E/Z*-isomers (Eq. (7)). Interestingly, substrates of this type can also undergo tandem 1,6–5,6-additions, indicating that the allenyl enolate formed by 1,6-cuprate addition is sufficiently electrophilic to react with another organometallic reagent in a carbometalation of the allenic double bond distal to the electron-releasing enolate moiety [25b].

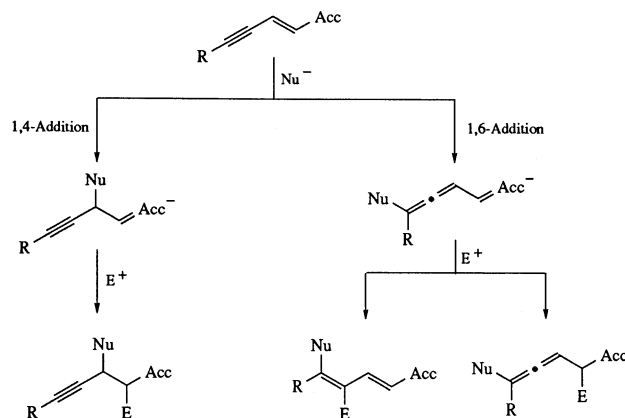


(7)

More interesting in preparative terms would be the possibility of shifting the regioselectivity of the electrophilic quenching reaction towards formation of allenes, since the number of synthetic methods for the preparation of functionalized allenes has been rather limited [26]. Furthermore, a stereoselective reaction of

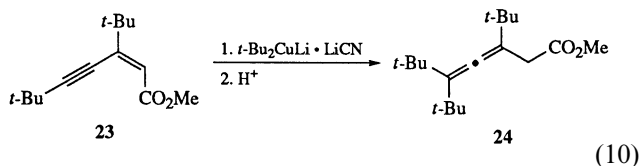
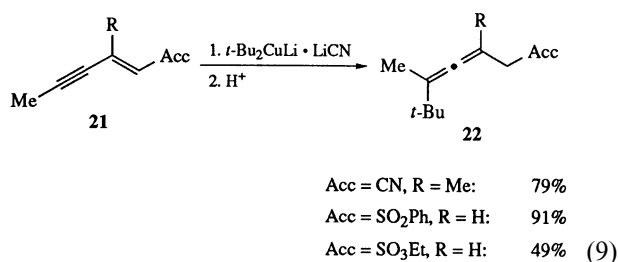
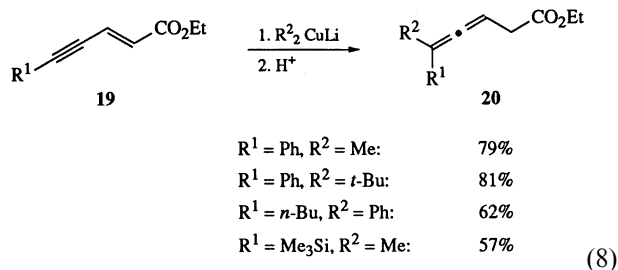
2.2. Acceptor-substituted enynes

Michael acceptors bearing an acceptor-substituted triple bond in conjugation with one or more double bonds react with organocuprates in a 1,4-addition mode exclusively [23]; 1,6-addition reactions, which would provide allenes after electrophilic capture, were not observed. In contrast to these transformations, nucleophilic additions to enynes with an acceptor substituent at the double bond are highly rewarding both



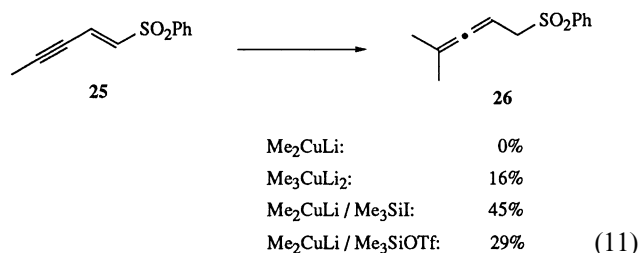
Scheme 2.

this type would open up a route to these axially chiral compounds in enantiomerically enriched or pure form. Indeed, the Gilman cuprate $\text{Me}_2\text{CuLi}\cdot\text{LiI}$ and cyanocuprates $\text{R}_2\text{CuLi}\cdot\text{LiCN}$ ($\text{R} \neq \text{Me}$) in diethyl ether react regioselectively with variously substituted 2-en-4-ynoates **19** in a 1,6-fashion (Eq. (8)). Protonation with dilute sulfuric acid gives the β -allenic esters **20** with alkyl, alkenyl, aryl, and silyl substituents in good yield [27]. Similarly, enynes **21** containing thioester, lactone, and dioxanone as well as keto, sulfonyl, sulfinyl, cyano, and oxazolidino groups react in a 1,6-manner to give the corresponding functionalized allenes **22** (Eq. (9)); only 1-nitro-1-en-3-yne is attacked at the $\text{C}=\text{C}$ double bond with the formation of 1,4-adducts [27c]. The differences in reactivity can be described qualitatively by the following reactivity scale: Acceptor (Acc) = $\text{NO}_2 > \text{COR}, \text{CO}_2\text{R}, \text{COSR} > \text{CN}, \text{SO}_3\text{R}, \text{oxazolidino} > \text{SO}_2\text{R} > \text{SOR} \gg \text{CONR}_2$. Remarkably, the regioselectivity of the cuprate addition to acceptor-substituted enynes is also insensitive to the steric properties of the substrate; enynes with *t*-butyl substituents at the triple bond (e.g. **23**) undergo 1,6-addition, even when the cuprate itself is sterically demanding (Eq. (10)) [27b]. The reaction is therefore highly suitable for the preparation of sterically encumbered allenes of type **24**.

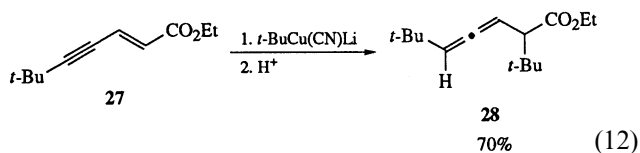


In order to achieve acceptable yields with less reactive Michael acceptors, it is often necessary to use more reactive organocopper reagents or Lewis acid activation. Thus, the reaction of enyne sulfone **25** with 5 equiv. of Me_2CuLi alone gave no trace of addition product, whereas the analogous reaction with Me_3CuLi_2 provided the desired allene **26** in 16% yield

(Eq. (11)) [27c]. With two equivalents of Me_2CuLi in the presence of one equivalent of Me_3SiI the yield increased to 45%, while with Me_3SiOTf the allene **26** was isolated in 29% yield. Only enyne amides fail to form 1,6-adducts under these conditions.

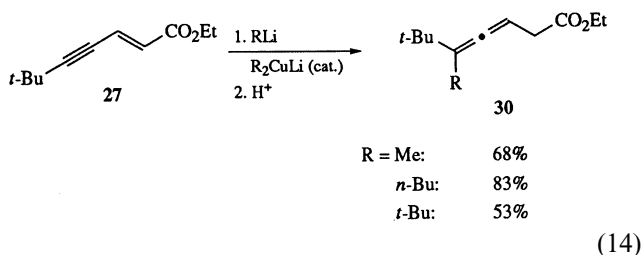
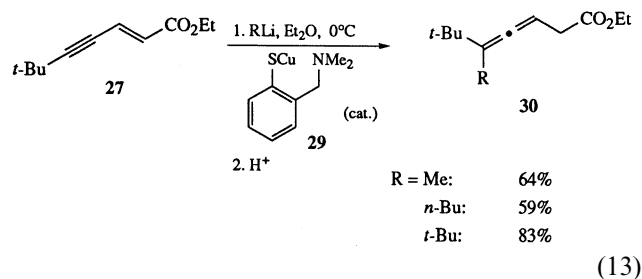


In contrast to the substrate, the organocuprate has a pronounced influence on the regioselectivity of the addition to acceptor-substituted enynes. While the Gilman cuprates $\text{Me}_2\text{CuLi}\cdot\text{LiI}$ and $\text{R}_2\text{CuLi}\cdot\text{LiCN}$ ($\text{R} \neq \text{Me}$) add regioselectively in a 1,6-manner, the Yamamoto reagent $\text{RCu}\cdot\text{BF}_3$ [3f] and the reagent combination $\text{RCu}/\text{Me}_3\text{SiI}$ [28] lead to 1,4-adducts [30]. In some cases, 1,4- and 1,6-reduction products are observed which may be formed by electron transfer from the cuprate to the substrate or by hydrolysis of a stable copper(III) intermediate [24,29]. So-called ‘lower order cyanocuprates’ $\text{RCu}(\text{CN})\text{Li}$ again show a different behavior: although they usually do not react with acceptor-substituted enynes, the cuprate $t\text{-BuCu}(\text{CN})\text{Li}$ undergoes *anti*-Michael additions with 2-en-4-ynoates and nitriles; e.g. the product **28** was obtained with 70% yield from enynoate **27** (Eq. (12)) [30]. The mechanistic aspects of this very unusual reaction are unknown; radical intermediates and electron transfer steps have not been found.



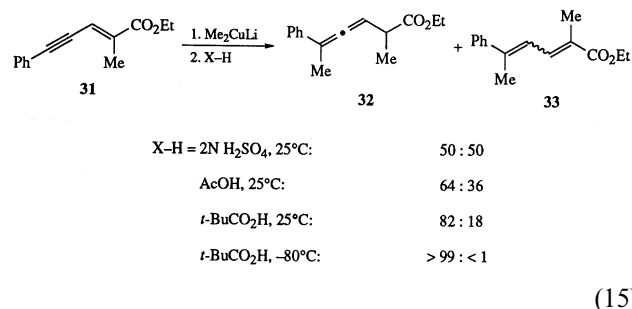
Similar to copper-catalyzed Michael additions of Grignard reagents to enones and activated dienes, the 1,6-addition to acceptor-substituted enynes can also be conducted under catalytic conditions. However, only very carefully controlled reaction conditions lead to the 1,6-adduct as the major product, i.e. use of copper (2-dimethylaminomethyl)thiophenolate (**29**) as catalyst and simultaneous addition of the substrate (e.g. **27**) and an organolithium reagent to a suspension of the catalyst **29** in diethyl ether at 0°C (Eq. (13)) [31]. Under these conditions variously substituted β -allenylcarboxylates **30** are obtained with yields comparable to those of the stoichiometric cases. Other copper(I) salts and the use of Grignard reagents as the nucleophile led to very low yields of 1,6-addition products. A second catalytic version takes advantage of the fact that the products of

the (stoichiometric) 1,6-cuprate addition, the lithium allenyl enolate and the organocopper compound, are formed as independent species. The cuprate can therefore be regenerated by addition of one equivalent of RLi such that it reacts with a further equivalent of the Michael acceptor. This procedure can, in principle, be repeated infinitely. The reaction is best conducted in a continuous mode by adding the substrate **27** and the organolithium reagent simultaneously to catalytic amounts of the cuprate (Eq. (14)) [30].

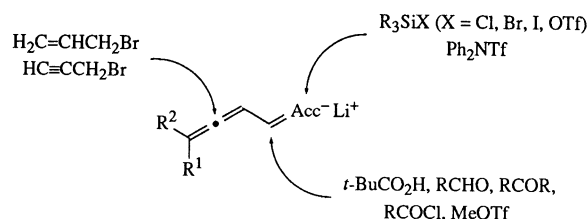


As mentioned repeatedly, a precondition for the successful preparation of allenes by 1,6-addition is that the allenyl enolate reacts regioselectively with an electrophile at C-2 (or at the enolate oxygen atom to give an allenylketene acetal; see Scheme 2). The regioselectivity of the simplest trapping reaction, the protonation of the allenyl enolate, depends on the steric and electronic properties of the substrate and the proton source. Whereas the allenyl enolates obtained from 3-alkynyl-2-cycloalkenones always provide conjugated dienones by protonation at C-4 (possibly via allenyl enols; see Eq. (7)) [25], ester enolates are usually protonated at C-2 (Eq. (8)), in particular when sterically demanding groups at C-5 block the attack of a proton at C-4 (Eq. (10)) [30,27]. However, with a substituent at C-2 of the enolate (e.g. the enolate formed from enyne **31** and Me_2CuLi), mixtures of allenes and conjugated dienes are formed, since now protonation at C-2 is sterically hindered. In the case of ester enolates this problem can be solved by using weak organic acids as a proton source; in particular pivalic acid (2,2-dimethylpropionic acid; see Eq. (15)): at room temperature, a ratio of **32**:**33** = 82:18 was observed, while at -80°C only the desired allene **32** was formed [27a]. In contrast to protonation, the regioselectivity of the reaction of other electrophiles with allenyl enolates derived from 2-en-4-ynoates is independent of the steric and elec-

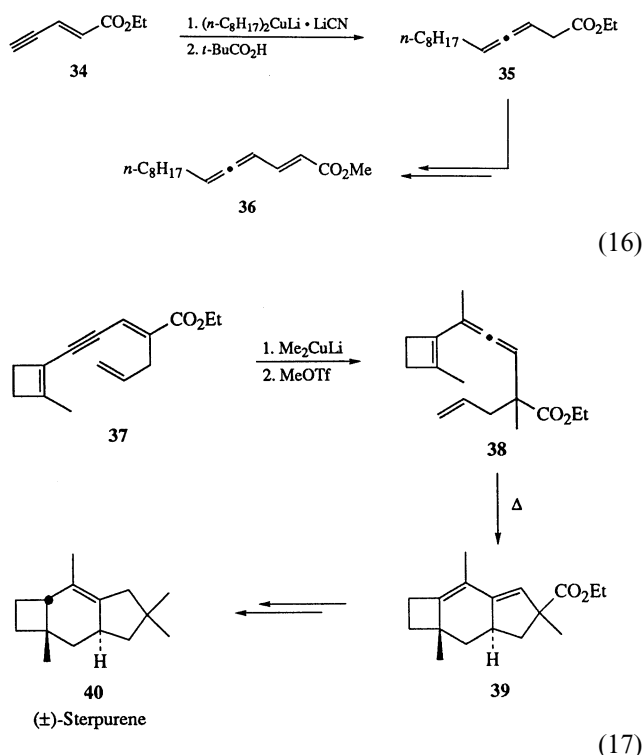
tronic properties of the reaction partners (Scheme 3) [30,32]. Hard electrophiles such as silyl halides and triflates react at the enolate oxygen atom to form allenylketene acetals, while soft electrophiles such as carbonyl compounds attack at C-2. Only allylic and propargylic halides react regioselectively at C-4 of the allenyl enolate to give substituted conjugated dienes. Again, cyclic allenyl enolates formed by cuprate addition to 3-alkynyl-2-cycloalkenones **17** show a deviating behavior: treatment with iodomethane provided product mixtures derived from attack of the electrophile at C-2 and C-4, and the reaction with aldehydes and silyl halides took place at C-4 exclusively [24,33].



The synthesis of allenes by 1,6-addition of organocopper reagents to acceptor-substituted enynes has found a wide range of preparative applications. In addition to sterically encumbered allenes (Eq. (10)) [27b] and simple terpenes such as pseudoionone [27a], allenic natural products can be prepared by this method (Eq. (16)) [30]. For example, 1,6-addition of lithium di-*n*-octylcuprate to enynoate **34**, followed by regioselective protonation with pivalic acid, yielded the allene **35** which was converted easily into the insect pheromone methyl 2,4,5-tetradecatrienoate (**36**). Another application of the 1,6-addition in natural product synthesis relies on vinylallenes as diene components in the Diels–Alder reaction (Eq. (17)). The synthesis of the fungal metabolite ($\pm$)-sterpurene (**40**) started with a 1,6-addition of lithium dimethylcuprate to an enynoate **37** and regioselective trapping with methyl triflate [34]. The vinylallene **38** thus formed underwent an intramolecular [4 + 2]-cycloaddition upon brief heating in toluene, and the tricyclic product **39** was converted into ($\pm$)-sterpurene (**40**) and also into several oxygenated metabolites.

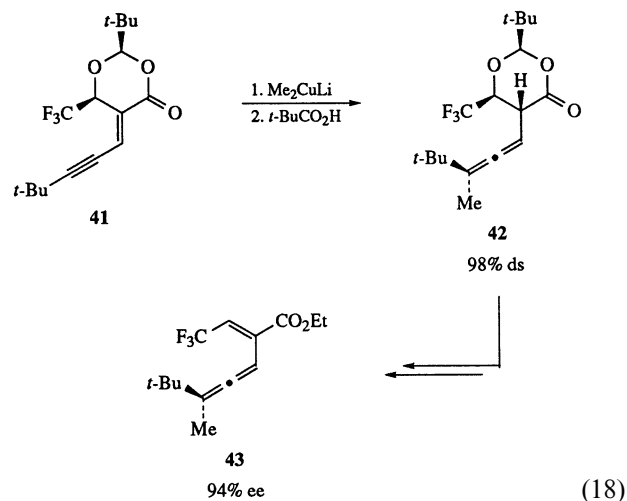


Scheme 3.

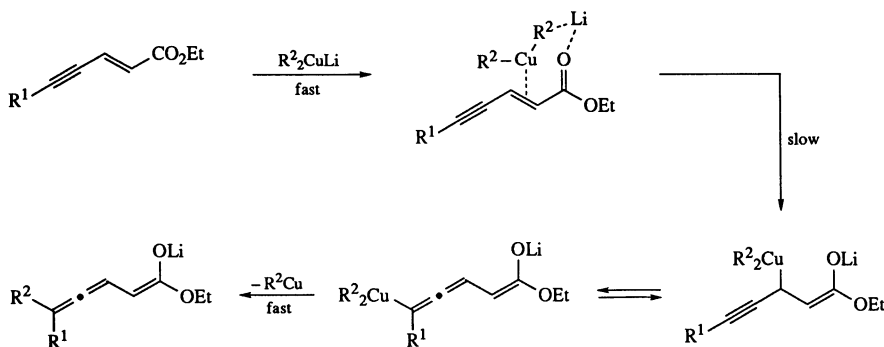


This intramolecular Diels–Alder reaction is an example of how axially chiral allenes, accessible through 1,6-addition, can be utilized to form new stereogenic centers selectively. This is also possible by intermolecular Diels–Alder reactions of vinylallenes [35], aldol reactions of allenyl enolates [36] and Ireland–Claisen rearrangements of silyl allenylketene acetals [37]. In order to access the required allenes in enantiomerically enriched or pure form, the 1,6-cuprate addition has to be conducted not only regio-, but also diastereo- or enantioselectively. This goal can be achieved by employing chiral 5-alkynylidene-1,3-dioxan-4-ones of type **41** as Michael acceptor; here, the equatorial *t*-butyl group forces the molecule to adopt a very rigid conformation and the trifluoromethyl group protects the top face of the enyne unit, making the preferred point of attack the underside of the molecule (Eq. (18)) [30,38].

Consequently, reaction with lithium dimethylcuprate and pivalic acid gave the desired allene **42** with a diastereoselectivity of 98% ds, and the stereochemical information generated in this step remained intact even after further conversion into a chiral vinylallene **43**.



In contrast to nucleophilic addition reactions to activated dienes, the mechanism of 1,6-cuprate additions to acceptor-substituted enynes is quite well understood, the main tools being kinetic and NMR spectroscopic investigations [30]. ¹³C NMR spectroscopic studies have revealed that these transformations proceed via π -complexes with an interaction between the π -system of the C=C double bond and the nucleophilic copper atom (a soft–soft interaction in terms of the HSAB principle), as well as a second interaction between the hard lithium ion of the cuprate and the hard carbonyl oxygen atom (Scheme 4) [39]. The use of ¹³C labeled substrates has confirmed that the cuprate does not interact with the triple bond, and it has also shed light on the structure of the metal-containing part of the π -complexes. Further intermediates on the way from the π -complex to the allenyl enolate could not be detected spectroscopically; however, kinetic measurements have revealed that an intramolecular rearrangement of the π -complex occurs in the rate-determining

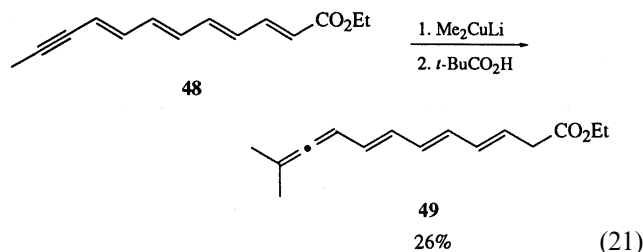
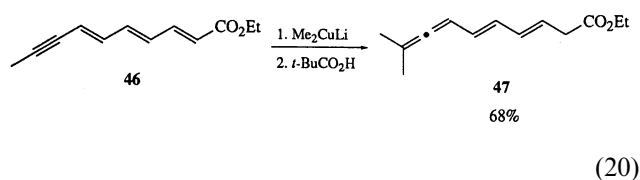
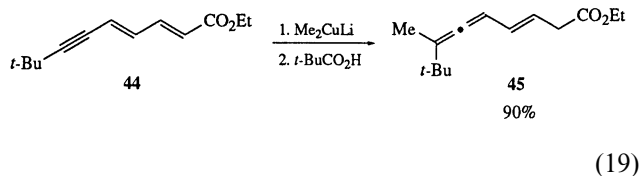


Scheme 4.

step [40]. These experimental results can be explained by assuming that a σ -copper(III) species is formed which could be in equilibrium with an allenic copper(III) intermediate. Both intermediates can undergo reductive elimination to produce the 1,4- and 1,6-adduct, respectively. The experimental result of exclusive formation of the 1,6-addition product may indicate that the hypothetical equilibrium lies on the side of the allenic copper(III) species, or that the reductive elimination of the latter occurs much faster than from the first intermediate.

2.3. Acceptor-substituted polyenyne

When one considers the high regioselectivity of the addition of organocuprates to acceptor-substituted enynes, it is natural to ask whether the preference of these reagents for triple bonds persists when the distance between the acceptor group and the triple bond is increased by the introduction of further C=C bonds. Of course, the number of possible regioisomeric products increases with increasing length of the Michael acceptor; for example, the 2,4-dien-6-ynoate **44** can be attacked by an organocopper reagent at C-3, C-5 or C-7, and the last possibility leads to a vinylogous allenyl enolate having four reactive positions (enolate oxygen, C-2, C-4, C-6). Therefore the high regioselectivity of the reaction of **44** with lithium dimethylcuprate is particularly remarkable: the cuprate attacks the triple bond exclusively and protonation with pivalic acid occurs at C-2 of the enolate, giving the 1,8-adduct **45** as the only isolable regioisomer with 90% yield (Eq. (19)) [35]. Analogously, the trienynoate **46** reacted in a 1,10-fashion to give the 3,5,7,8-tetraenoate **47** (Eq. (20)), and even the 1,12-adduct **49** could be obtained from the Michael acceptor **48** which contains four double bonds between the triple bond and the acceptor substituent (Eq. (21)). In the latter case, however, the yield was only 26%; this is probably due to the reduced stability of the addition products with increasing length of the conjugated π -system (the 1,12-adduct was the only isolable reaction product, apart from polymeric compounds) [30,35].

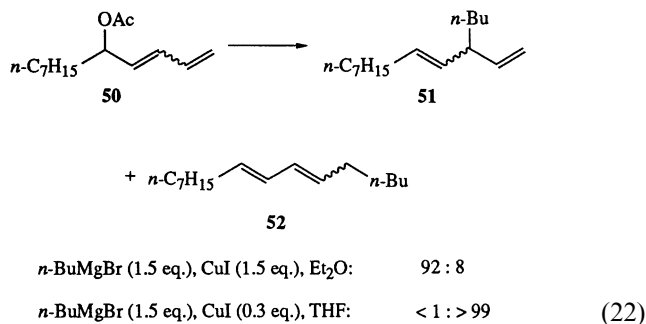


These transformations and those summarized in the previous section make clear that Michael acceptors containing any combination of double and triple bonds undergo regioselective addition reactions with organo-copper reagents. The following rule holds: 'Michael acceptors with any given arrangement of conjugated double and triple bonds react regioselectively with organocuprates at the triple bond closest to the acceptor substituent'. Similar to the 1,6-cuprate addition to acceptor-substituted enynes (Scheme 4), these reactions start with the formation of a cuprate π -complex at the double bond neighboring the acceptor group [39] and may then proceed via an allenic σ -copper(III) intermediate which produces the addition product by reductive elimination of RCu [30].

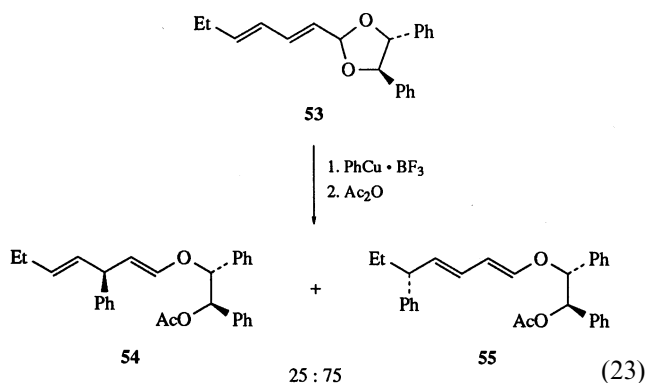
3. Copper-mediated substitution reactions of extended substrates

Compared to addition reactions of extended Michael acceptors, only a few examples of copper-mediated substitutions of extended electrophiles have been reported to date. Goering and coworkers [41] studied substitution reactions of various dienyl carboxylates with organocuprates (and Grignard reagents in the presence of catalytic amounts of copper salts) and found that the ratio of the three possible regioisomers (i.e. α -, γ - and ϵ -alkylated product) depends strongly on the substrate and reaction conditions. In the case of dienyl acetate **50**, Utimoto and coworkers [42] were able to shift the regioselectivity of the substitution from the γ - to the ϵ -product by variation of the solvent and amount of copper salt (Eq. (22)): when the reaction is carried out in diethyl ether with *n*-BuMgBr and stoichiometric quantities of CuI, the S_N2' -(1,3)-substitution product **51** is observed as major product, whereas with catalytic quantities of CuI and THF as solvent the conjugated diene **52** is formed exclusively (i.e. S_N2'' -(1,5)-substitution takes place under these conditions). The dependence of the reaction course on the *n*-BuMgBr:CuI ratio lends credence to the postulate that different organocopper species are responsible for the formation of the regioisomeric products. With equimolar amounts of Grignard reagent and copper salt the probable active species is the monoalkylcopper compound *n*-BuCu·MgBrI, which leads to **51**. By contrast, an excess of Grignard reagent should provide the mag-

nesium cuprate $n\text{-Bu}_2\text{CuMgBr}$ as the reactive species, which gives the 1,5-substitution product **52**.

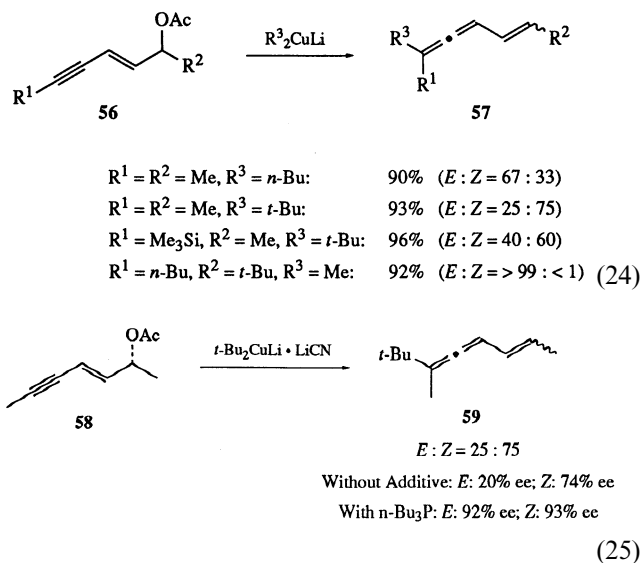


Substitution reactions of chiral dienyl electrophiles can also be carried out diastereo- and enantioselectively. For example, $\text{S}_{\text{N}}2'-(1,3)$ -substitutions of dienyl carbonates [43] were reported to proceed with high *anti*-selectivity (this stereochemical course is often observed in copper-mediated $\text{S}_{\text{N}}2'$ reactions of simple allyl electrophiles [3]). An interesting observation was recently made in the reaction of chiral dienyl acetal **53** with the Yamamoto reagent $\text{PhCu} \cdot \text{BF}_3$, which leads to a 1:3 mixture of substitution products **54** and **55** (Eq. (23)): whereas the former is the result of an *anti*- $\text{S}_{\text{N}}2'$ -substitution, the $\text{S}_{\text{N}}2'-(1,5)$ -substitution product **55** is formed *syn*-selectively [44]. A mechanistic rationalization of this surprising result has not yet been put forward.



The corresponding copper-mediated $\text{S}_{\text{N}}2'-(1,5)$ -substitution reactions of conjugated enyne acetates **56** take place with high regioselectivity and furnish vinylallenes **57** with variable substitution patterns (Eq. (24)) [45]. Although the substitution products were often obtained as mixtures of *E/Z* isomers, complete stereoselection with regard to the olefinic double bond of the vinylallene could be achieved in some cases. These 1,5-substitutions can also be applied to enyne oxiranes as substrates; additionally, they can also be performed without loss of efficiency under catalytic conditions, i.e. by simultaneous addition of RLi and the substrate to catalytic amounts of the cuprate [45]. Fortunately, these transformations take place with high enantioselectivity

when appropriate reaction conditions are used (Eq. (25)). For example, treatment of the enantiomerically pure enyne acetate **58** with $t\text{-Bu}_2\text{CuLi} \cdot \text{LiCN}$ in diethyl ether at -90°C provided vinylallene **59** as a 1:3 mixture of *E/Z*-isomers, and GC analysis revealed enantiomeric excesses of 20% ee for (*E*)-**59** and 74% ee for (*Z*)-**59** [45]. The loss of stereochemical information is probably due to the racemization of the allene by the cuprate. According to Alexakis et al. [46], this racemization can be prevented by addition of phosphines to the mixture which act as a strong ligand to copper. In fact, vinylallene **59** was obtained with enantiomeric excesses of 92% ee for the *E*-isomer and 93% ee for the *Z*-isomer when the reaction was carried out in the presence of $n\text{-Bu}_3\text{P}$. Thus, the 1,5-substitution of enyne acetates constitutes one of very few cases of efficient remote stereocontrol in organocopper chemistry.



4. Concluding remarks

Organocopper reagents have been utilized with great success in organic synthesis in the last 30 years. The more recent examples discussed in this review document the excellent performance of these organometallic compounds in regio- and stereoselective transformations of substrates with extended π -networks. Significant new developments include 1,6-, 1,8-, 1,10-, and 1,12-additions, as well as 1,5-substitution reactions. These new transformations not only proceed with high regio- and stereoselectivity and can therefore be utilized in target-oriented synthesis, but they also provide new information about the mechanism of copper-mediated carbon–carbon bond forming reactions. Future research in this area will almost certainly concentrate on the further development of new catalytic and enantioselective transformations.

Acknowledgements

The work on copper-mediated addition and substitution reactions was sponsored by the Deutsche Forschungsgemeinschaft, the Volkswagen-Stiftung, the Fonds der Chemischen Industrie, the European Community, and the Ministerium für Wissenschaft und Forschung des Landes Nordrhein-Westfalen. We also thank all coworkers whose names are listed in the references for their contributions.

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